Reinforcement Learning Foundations

Block 1 Day 1

Semester 1, 2026–2027 — Teaching Week 3

Session Overview

Session objective: Build intuition for trial-and-error learning, then move into update rules, delayed reward, and safe exploration choices.
Timing target (min): 69
Topic anchors: reinforcement learning; q-learning; sarsa; delayed reward; exploration; safety
Padlet board: https://padlet.com/qmul/b1-d1-3o42dysklnb5eeyd

Exercises

Q1 Starter Intuition

Question type Concept → **Working mode** Pair → **Expected time** 8 min

Prompt

Start with the question from the lecture opener: How do babies learn to walk without step-by-step instructions? Use it to explain what trial and error, feedback, and improvement over time mean in reinforcement learning.

Task details

- **Type:** concept
- **Mode:** pair
- **Time:** 8 min
- **Deliverable:** short explanation
- **Assessment focus:** connect everyday trial-and-error learning to the core RL idea
- **Expected length:** 3 bullets + 1 example

How to submit

Post one pair Padlet comment with three short bullets and one everyday example besides walking.

Discussion in class

Which part of this story is most similar to delayed reward?

Why this helps

Useful for short introductory questions that ask what reinforcement learning is trying to do.

References

- Russell & Norvig (2021) AIMA 4e, Ch.22, pp.789-822
- Poole & Mackworth (2023) 3e, Ch.13, pp.583-608

Q2 Core Concept Check

Question type Concept	→	Working mode Pair	→	Expected time 10 min

Prompt

Compare reinforcement learning and supervised learning using the three lecture ideas of feedback timing, how training data is collected, and what objective is optimised. Finish with one real example where reinforcement learning is the better fit.

Task details

- **Type:** concept
- **Mode:** pair
- **Time:** 10 min
- **Deliverable:** comparison bullets
- **Assessment focus:** distinguish reinforcement learning from supervised learning using day-1 concepts
- **Expected length:** 4 bullets + 1 example

How to submit

Post one pair Padlet comment with one bullet for each contrast and one example.

Discussion in class

Which contrast matters most when the agent must act in the real world?

Why this helps

Useful for theory questions that ask for a direct comparison between learning paradigms.

References

- Russell & Norvig (2021) AIMA 4e, Ch.22, pp.789-822
- Poole & Mackworth (2023) 3e, Ch.13, pp.583-608

Q3 Structured Problem

Question type Problem	→	Working mode Individual	→	Expected time 12 min

Prompt

In a smart-greenhouse control task, the agent is in state g4, takes action Heat, receives reward -1, moves to state g5, and currently has Q(g4, Heat)=2.4, max_a Q(g5,a)=6.0, alpha=0.5, gamma=0.8. Identify the state, action, reward, and next state, then compute the updated Q(g4, Heat).

Formula reminder

$$Q_{\text{new}}(s,a) = Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

Use this when the task asks for one update step.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 12 min
- **Deliverable:** worked update
- **Assessment focus:** identify RL elements correctly and execute one Q-learning update
- **Expected length:** 4 labels + 4 calculation lines

How to submit

Post your own Padlet comment showing the four RL elements, the substitution into the update rule, the target, and the final value.

Discussion in class

Which quantity in the update says what the agent expects from the next state?

Why this helps

Direct practice for state-action-reward identification and Q-learning calculations.

References

- Russell & Norvig (2021) AIMA 4e, Ch.22, pp.789-822
- Poole & Mackworth (2023) 3e, Ch.13, pp.583-608

Q4 Stretch Extension

Question type Stretch → **Working mode** Pair → **Expected time** 12 min

Prompt

Using the lecture ideas about reward signals, delayed reward, learning rate, and discounting, answer these four linked questions: What happens to Q-values if rewards are always 0? What happens if rewards only arrive after many steps? What does a larger alpha change? What does a larger gamma change?

Task details

- **Type:** stretch
- **Mode:** pair
- **Time:** 12 min
- **Deliverable:** short analytical note
- **Assessment focus:** reason about zero reward, delayed reward, and the roles of alpha and gamma
- **Expected length:** 4 numbered answers

How to submit

Post one pair Padlet comment with four numbered answers in simple language.

Discussion in class

Which setting makes learning look slow even when the update rule is correct?

Why this helps

Good extension for explanation questions about delayed reward, discounting, and learning-rate effects.

References

- Russell & Norvig (2021) AIMA 4e, Ch.22, pp.789-822
- Poole & Mackworth (2023) 3e, Ch.13, pp.583-608

Q5 Collaborative Mini-Case

Question type	Working mode	Expected time
Case	Group	15 min

Prompt

Imagine a tea-serving robot that learns by trial and error. Design a reward function that encourages task completion, avoids spills, avoids burns, and avoids wasting ingredients. Then add two controls that reduce reward hacking.

Task details

- **Type:** case
- **Mode:** group
- **Time:** 15 min
- **Deliverable:** reward-design sketch
- **Assessment focus:** design a reward function with both task and safety terms
- **Expected length:** 4 reward terms + 2 anti-hacking controls

How to submit

Post one group Padlet comment with four reward terms, their signs or weights, and two anti-hacking controls.

Discussion in class

Which unsafe shortcut would your robot still try first if the reward is badly designed?

Why this helps

Supports design questions on reward specification, safety, and unintended behaviour.

References

- Russell & Norvig (2021) AIMA 4e, Ch.22, pp.789-822
- Poole & Mackworth (2023) 3e, Ch.13, pp.583-608

Q6 Exam-Style Constructed Response

Question type Exam → **Working mode** Pair → **Expected time** 12 min

Prompt

A hospital cleaning robot first learns in simulation and then fine-tunes in a real corridor where patients and staff may appear unexpectedly. Recommend a safer training setup for the real-world stage. Your answer must say whether learning should continue online or switch to evaluation-only deployment, and it must include one exploration control and one stop condition that would halt unsafe behaviour.

Task details

- **Type:** exam
- **Mode:** pair
- **Time:** 12 min
- **Deliverable:** claim-evidence-justification response
- **Assessment focus:** justify a safer RL training setup under noise and human-safety constraints
- **Expected length:** 180-240 words

How to submit

Post one pair Padlet comment with a clear recommendation, two technical reasons, one exploration control, and one stop condition.

Discussion in class

Which safety control matters more in the real corridor: limiting exploration or stopping unsafe episodes quickly?

Why this helps

Matches longer answers that ask for training trade-offs, deployment safety, and practical RL guardrails.

References

- Russell & Norvig (2021) AIMA 4e, Ch.22, pp.789-822
- Poole & Mackworth (2023) 3e, Ch.13, pp.583-608

Submission Instructions

Use the seeded Padlet posts for Q1–Q6. Q3 is individual. Q1, Q2, Q4, and Q6 are pair tasks. Q5 should be one joint group comment. Start each comment with your names or group label.

Whole-Class Debrief Prompts

- Which part of everyday trial-and-error learning maps most directly to reinforcement learning?

- What makes delayed reward harder than immediate reward?
- Which reward-design mistake is most likely to cause unsafe behaviour?

Exam Transfer Note (Day-Level)

Maps to exam questions on RL intuition, delayed reward, Q-learning updates, parameter effects, reward design, and safety-aware method choice.